Water Resources Management (2026) 40:522
https://doi.org/10.1007/s11269-026-04883-x

The Application of Artificial Intelligence in Streamflow Forecasting: A Review

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Received: 17 October 2024 / Accepted: 25 August 2026 / Published online: 9 September 2026
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Abstract
Accurate streamflow modelling and forecasting are crucial for proper water resources planning, engineering, and management. Since the last two decades, Artificial Intelligence (AI) methods have been increasingly developed and used in streamflow forecasting due to their robustness in dealing with non-linear, uncertain, and complex hydrological datasets. The current paper is a retrospective study on the different AI methods used in streamflow forecasting. Based on the review of research papers published in different journals from 1999 to 2023, this paper presents an analysis of the various AI techniques and algorithms in use. It identified feed-forward neural network (FFNN) as one of the most prominently used AI methods in streamflow forecasting. Also, the review examined papers from the perspective of forecast window time frame used, only to find that majority of the research considered a short forecast window of 1-7 day(s) lead time as compared to longer time frames. Additionally, the paper also provides information on the different input parameters used by researchers over time in defining their AI models. The analysis shows that primarily used parameters include streamflow, rainfall, and temperature data, and negligible use of snow or other cyclic factors as inputs. Based on all these statistical information, this review paper contributes to the possibility of exploring other inputs parameters and extending the existing forecasting window time frame for better forecasting of streamflow, and hence, efficient management of water resources. It has also forwarded directions for future research in streamflow modelling, including the exploration of time series neural network models for forecasting streamflow over larger windows. This review thus, identifies the challenges associated with streamflow prediction and the possible ways to overcome them.

Keywords Streamflow · Artificial intelligence · Neural network · Prediction · Forecast

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1 Introduction

Streamflow and its prediction are fundamental to many aspects of water resource management for purposes such as irrigation, residential and commercial use, hydropower generation or environmental management (Sharma and Machiwal 2021). Sudden floods or prolonged droughts can jeopardize the planning and management of water for various activities. Being able to predict streamflow to manage those types of events, inform, or optimize operational decisions has significant potential value. For instance, Elephant Butte Reservoir, a multi-objective reservoir, provides electrical power and water for south-central New Mexico and West Texas, including irrigation water for 68,708.25 ha (169, 650 acres) of farmland. Caballo Reservoir, located 40.25 km (25 miles) downstream of Elephant Butte Reservoir, is primarily fed by the water released from Elephant Butte Reservoir and provides direct release of irrigation water downstream into the Rio Grande Project during the primary cropping season (Sabzi et al. 2018).

A reliable release plan from Caballo Reservoir to meet the existing water demands downstream of the Rio Grande Project is directly dependent on the storage level of Caballo Reservoir, whose optimal operation plan must be developed under uncertain conditions. Moreover, providing optimal storage levels in both reservoirs can minimize the total evaporation loss from both reservoirs (Sabzi et al. 2018). The development of the release plan is dependent on a highly accurate long term streamflow forecasting system. It will enable stakeholders to plan and manage the operation of Caballo Reservoir well in advance. This will not only result in smooth operation of the reservoir, but will also result in minimum water loss from the reservoir through evaporation (Sabzi et al. 2018). Streamflow forecasting is great importance in increased hydropower generation. For example, a hybrid streamflow forecasting system was developed to predict streamflow 1-6 months ahead for two stations in Yellow river, China. The resultant improvement in hydro power operational decisions enhanced generation by up to 3.39 %. However, results showed that as forecast lead time was increased, accuracy of streamflow forecasts reduced and that in turn, affected hydro power generation, which consequently got reduced. This shows that accurate streamflow forecasting is extremely important when making operational decisions, such as those related to hydro power generation, where accurate forecasts can potentially result in increased hydro power generation and vice versa (Guo et al. 2021a).

Water resources hold great significance in the socio-economic development of a country (Feng et al. 2020c; Liu et al. 2017; Sun et al. 2018). In many countries, however, these resources are unevenly distributed in both space and time, and could even be unmatched with the economic development and water demand process (Bai et al. 2015; Feng et al. 2020b). For example, North China with around 19 % water resources, is required to support about half of gross domestic product, population, cultivated land and national territorial area, showing huge contradiction between water resources supply and healthy economic development (Lei et al. 2011; Su et al. 2018). However, streamflow is not an easily predictable parameter due to its dependence on many other variables, and varying spatio-temporal characteristics.

The hydrological components of streamflow, due to their high variability and complexity, make streamflow forecasting a challenging task. Accurate forecasting of streamflow is important for water resources management, especially in times of global climate change, and the associated extreme climate events. For instance, significant changes in water levels and flow patterns, the normal operation of reservoir networks can have adverse effects on a num-

ber of economic interests, i.e., irrigated agriculture, forestry, fisheries etc, and on the human population (Castelletti et al. 2008). For effective operation of water resources systems, it is important to forecast streamflow hours, days, months, or even longer ahead (Kişi 2011).

Streamflow forecasting models are evaluated using different metrics. Popular ones include the mean absolute error (MAE), root mean squared error (RMSE), coefficient of determination (R^2) and Nash–Sutcliffe model efficiency coefficient (NSE). Each has its own pros and cons depending on the relative scenario it is employed in. Evaluation metrics provide useful information on how well a predictive model is doing and areas where it could potentially be improved. A detailed discussion on the different evaluation metrics used in streamflow forecasting applications will be provided later in the study.

Streamflow behavior over various lead times is controlled by few important predictors. These include climate drivers like ENSO which are useful for forecasting over large lead times. Similarly, snow controlled parameters such as snow melt, snow density etc are also important in long lead time forecasting. Conversely, weather driven variables such as temperature and rainfall control streamflow behavior over short lead times.

Streamflow forecasting can be divided into two categories: (i) short-term forecasting, and (ii) long-term forecasting. Short-term forecasting covers a forecasting horizon ranging from the hourly to daily scale and is important for efficient operation of flood response systems. Long-term forecasting, on the other hand, covers a temporal scale ranging from weekly to annual forecasts. It is essential for the planning and operation of reservoirs, water releases, irrigation management related decisions, release of water for residential and commercial use, and several other purposes (Yaseen et al. 2015).

Different statistical forecasting models have been employed by researchers over the years to predict streamflow. These include the auto-regressive (AR), moving average (MA), auto-regressive integrated moving average (ARIMA) and auto-regressive moving average with exogenous input (ARIMAX), seasonal ARIMA (SARIMA) and seasonal ARIMAX (SARIMAX). Although these models have proven to be quite effective when working with time series data, in the case of complex and non-linear data, their performances have shown decay. However, the last three decades have witnessed a massive increase in the number of studies conducted on forecasting streamflow using AI techniques (Yaseen et al. 2015). AI models have proven effective in modelling non-linear and highly complex data, which are characteristics of any real-world data, let alone streamflow data. Deep learning and unsupervised learning have exhibited immense potential in modelling real world applications (Dean et al. 2012). Recent times have seen the transformation of machine learning (ML) as a powerful tool for technological advancement (Rolnick et al. 2022). AI models based on regression, time series analysis, mathematical optimization and classification etc, have been employed for streamflow forecasting on a regular basis. The AI methods that researchers have used over the last three decades for forecasting streamflow could be broadly classified into the following categories:

- *Machine learning models (regression, classification and time series based)*: these include linear regression (LR), artificial neural network (ANN), support vector machine (SVM), least squared-support vector regression (LS-SVR), extreme learning machine (ELM), convolutional neural network (CNN), long-short term memory (LSTM) and recurrent neural network (RNN) etc.
- *Evolutionary computation algorithms:* these include the genetic algorithm (GA), genetic programming (GP) and particle swarm optimization (PSO) etc.

- *Wavelet Conjunction models:* they are based on a signal processing tool and include the discrete wavelet transform (DWT) and continuous wavelet transform (CWT) algorithms.
- *Hybrid models:* these include models which could be a combination of machine learning models and one or more of evolutionary computation algorithms, fuzzy logic based systems (for example, ANFIS) and statistical models etc.

The most commonly used AI methods for streamflow forecasting include ANN, RNN, LSTM, ELM, SVM, and ANFIS. It is important to take hydrological errors into consideration when performing streamflow forecasting (McInerney et al. 2024). It is also important to confront uncertainties in streamflow modelling to achieve acceptable forecast results (Renard et al. 2010).

Input variable(s) selection is another aspect which has a big effect on performance of AI models (Noori et al. 2011; Snieder et al. 2020; Meng et al. 2021; Hadi et al. 2019; Dariane and Azimi 2018; Niu and Feng 2021; Maier et al. 2014; Bowden et al. 2005). Researchers have primarily been using input parameters like streamflow, rainfall, temperature, etc. while modelling these AI methods. A discussion on the applicability of other parameters, such as snowfall, wind speed and soil moisture etc, are still illusive. Snow in particular impacts streamflow greatly (Uysal et al. 2016b; WMO 1986; Gurtz et al. 2003; Berezowski et al. 2015). However, snow data availability is a major challenge making its applicability limited (Wrzesien et al. 2019; Crumley et al. 2020; Doesken and Robinson 2009; Slater et al. 2013). Wind speed on the other hand is known to impact streamflow indirectly through its effects on the behavior of rainfall (Chubb et al. 2015). An exploration into the possible influence of such input factors in streamflow forecasting is critical as it may affect the performance of an AI method.

Forecasting lead time is another interesting feature of any AI system and majority of the research are found to revolve around a short window time frame. A longer prediction time frame is critical for managing water resources efficiently. However, larger forecast lead times have adverse effects on model accuracy (Cheng et al. 2020; Schepen et al. 2016; Kalra and Ahmad 2009; Crochemore et al. 2016; Wyatt et al. 2020). As such, use of diverse input combinations and selection of forecast window are important highlights of this current review.

To fulfill these objectives, papers from different journals, which have been collected based on a 24-year time frame (1999 to 2023), are used. The review is performed on AI centred mechanisms with primary focus on ANN, RNN, CNN, LSTM, SVM, LS-SVM and ANFIS in streamflow modelling and forecasting.

The paper begins with a section describing the basic theory of each of these AI methods followed by a comprehensive review of prominent studies. The reviewed literature is analysed and evaluated in (Sections 2 and 4). Directions for future research are presented in (Section 5). Summary and conclusions are presented in the last part of the study (Section 6).

2 Artificial Intelligence Models Employed for Streamflow Forecasting

Different AI methods have been used over the years for streamflow forecasting. Figure 1 depicts the AI methods discussed in this study, while Table 1 (in the Appendix section) provides an overview of the AI method(s) used by each paper discussed in this study.

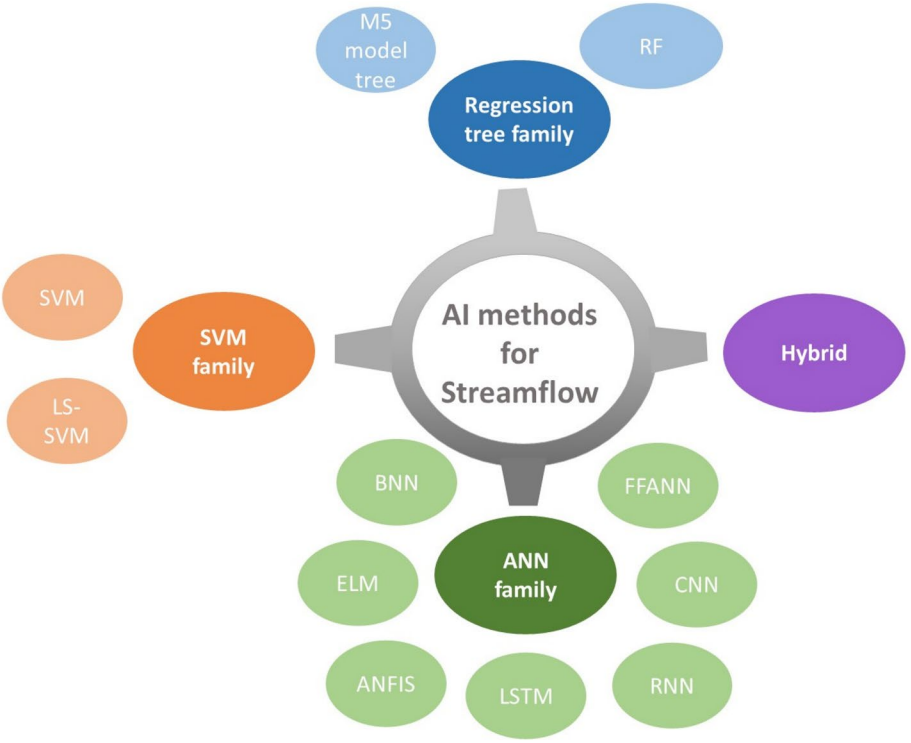


Fig. 1 AI methods discussed in this study

2.1 Artificial Neural Network (ANN) Family

An Artificial Neural Network (ANN) is a parallel processing system that resembles the neural network of the human brain (Haykin 1998). An ANN is composed of three layers: (i). an input layer, (ii). one or more hidden layer(s) and (iii). an output layer. Each layer contains a certain number of neurons or units depending on the problem at hand. The neurons are information processing units, where information from each of them is passed to each neuron in the succeeding layer and so on.

Like any other machine learning approach, ANNs convert a set of model inputs into an output via a complex mathematical relationship (Maier et al. 2023). Each neuron applies a non-linear transformation, called an activation function, to its net input in order to calculate its output (Committee 2000). Different variants of ANN have been employed by researchers for predicting streamflow over the years (Maier and Dandy 2000). Prominent ANN variants are discussed in following sub-sections.

2.1.1 Feed Forward ANN

A feed forward ANN is one in which information flows only in one directions, i.e., forward direction. There is no look back or feed back from the output layer back into the input layer. Figure 2 depicts the general architecture of a feed forward ANN.

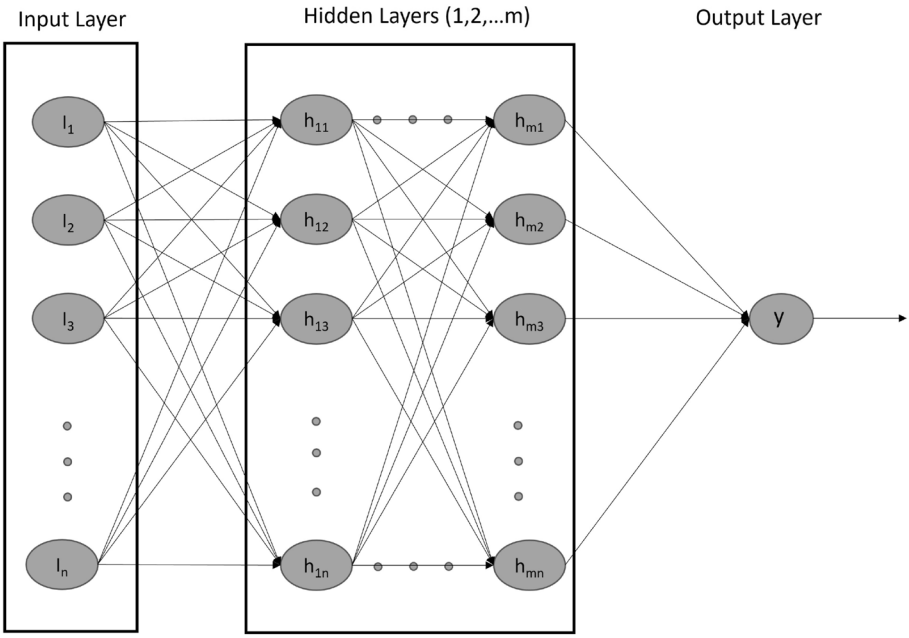


Fig. 2 General architecture of Artificial Neural Network (ANN) (Chouaibi 2022)

ANNs in general are able to model non-linearities in data well. Feed forward ANNs are no exception. This feature allows them to understand underlying relationships between different input variables as well as the relationships between input variable(s) and target variable better than standard regression methods such as linear regression. This quality has made the application of feed forwards ANNs very common in streamflow forecasting (Chitsaz et al. 2016; Selvi and Huseyinov 2020; Zheng et al. 2018; Cutore et al. 2007; Dolling and Varas 2002; Sharma et al. 2021; Besaw et al. 2010; Chokmani et al. 2008; Vidyarthi and Jain 2022; Can et al. 2012; Ahmed and Sarma 2007; Toth and Brath 2007; Wu et al. 2005; Fan et al. 2021; Cui et al. 2020).

As is evident from Fig. 2, the flow of information in a feed forward ANN in unidirectional, meaning that it does not incorporate feed back in its system. This feature makes it difficult for feed forward ANNs to predict multi-step ahead streamflow accurately. In fact, forecast accuracy of feed forward ANNs deteriorates rapidly as the prediction lead time is increased (Sun et al. 2019; Muluye and Coulibaly 2007).

On the other hand, feed forward ANNs are known to perform very well when forecasting streamflow one time step ahead (Sun et al. 2019; Chitsaz et al. 2016; Worland et al. 2019; Birikundavyi et al. 2002; Ali and Shahbaz 2020; De Faria et al. 2022; Dehghani et al. 2014; Makkeasorn et al. 2008; Sattari et al. 2012; Choudhury et al. 2018; Uysal et al. 2016a; Wang et al. 2019). This, as discussed previously, is due to the ability of feed forward ANNs to model non-linearities in data well, allowing them to generalize on data better.

A draw back with using feed forward ANN for streamflow forecasting is its inability to model peak flows accurately as is reported in various case studies (Parasuraman et al. 2006; Gunathilake et al. 2021; Chiang et al. 2004; Hsu et al. 2002; Danandeh Mehr et al. 2015;

Moradkhani et al. 2004; Cigizoglu 2005; Hu et al. 2005; Yaseen et al. 2016a). This could potentially be down to a lack of immunity in feed forward ANNs to sudden fluctuations in data which in turn is down to the absence of feed back mechanisms in feed forwards ANNs, depriving them of maintaining memory for establishing context.

2.1.2 Less Common ANNs (CNN, RNN)

Convolutional neural network (CNN) is a type of feed forward neural network capable of extracting features from data with convolution structures (Li et al. 2021a). This quality makes it a useful tool in streamflow forecasting. In particular, CNN is quite useful in optimal feature selection for modeling streamflow. Figure 3 shows the general architecture of CNN.

The ability of CNN to extract important information from predictors has made it useful in streamflow forecasting applications (Apaydin et al. 2021; Shu et al. 2021). Since streamflow behavior is largely dependent on several other variables, CNN could prove to be a useful tool in extracting useful information from those variables that would aid in enhanced streamflow forecast accuracy.

Recurrent neural network (RNN) is a modified version of the conventional feed forward ANN in that it consists of a recurrent or feedback mechanism enabling it to retain previous information (Chiang et al. 2017). This allows an RNN to effectively model time series data. Figure 4 depicts the general architecture of RNN.

The feed back feature in RNN allows it to forecast sequential data accurately as compared to feed forward ANN. For example, in the case of forecasting peak flow, feed forward ANN was reported to have struggled in predicting it accurately. RNN could potentially be employed to solve this problem (Chiang et al. 2017).

Since RNN doesn't maintain long term memory, it struggles to predict streamflow accurately over large lead times. This lead to the development of an improved version of the RNN called LSTM, that will be discussed in the following sub-section.

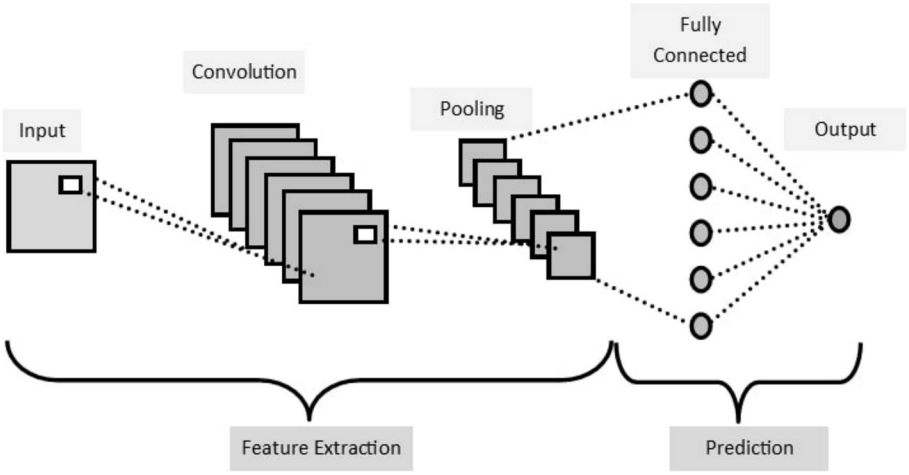


Fig. 3 General architecture of Convolutional Neural Network (CNN) (Alzubaidi et al. 2021)

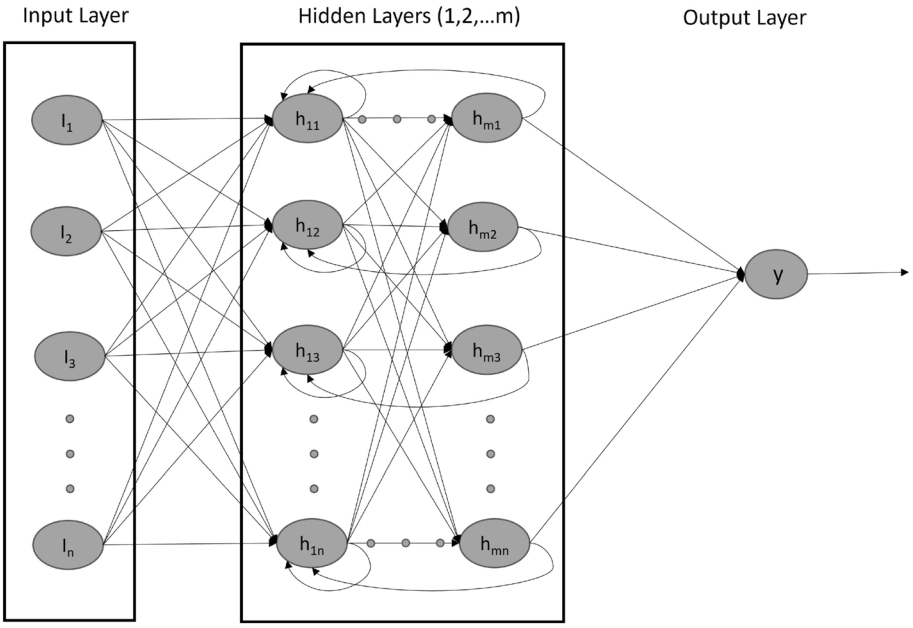


Fig. 4 General architecture of Recurrent Neural Network (RNN) (Siłka et al. 2022)

2.1.3 Long Short-Term Memory (LSTM)

LSTM is perhaps the most widely used model in forecasting problems involving time series. This is due to the fact that LSTM is very well equipped in modelling long term temporal relationships in data, as well as addressing the gradient vanishing problem issues that the conventional RNN model could not handle well (Ni et al. 2020). Figure 5 shows the general architecture of LSTM. LSTM consists of gates that control the flow of information to and from the model, making them core elements of the LSTM architecture. The first gate is the forget gate, which decides what information to remove from the cell state. A value of 1 represents completely keeping the information while a 0 represents completely removing it.

The next step involves making decision on what new information should be stored in the cell state. Here, the input gate decides which information is utilized for updating the cell state, while a non-linear activation function creates a fresh value that could be added to the cell state.

The cell state is then updated. The output gate is responsible for controlling the flow of information from cell state into the new hidden state. At the end, the new hidden state is computed.

LSTM is known to model sequential modeling problems better than most of its deep learning counter parts because of its ability to maintain useful memory, allowing it to develop context when forecasting complex variables (Apaydin et al. 2021). This feature allows LSTM to forecast streamflow with a higher degree of accuracy as compared to other predictive models (Huang et al. 2021).

The feed back system and ability to maintain long term memory in LSTM aids it in forecasting streamflow accurately multiple time steps ahead (Asaad et al. 2022; Cheng et al. 2020; Wang et al. 2022). This makes LSTM superior to most predictive systems as it helps

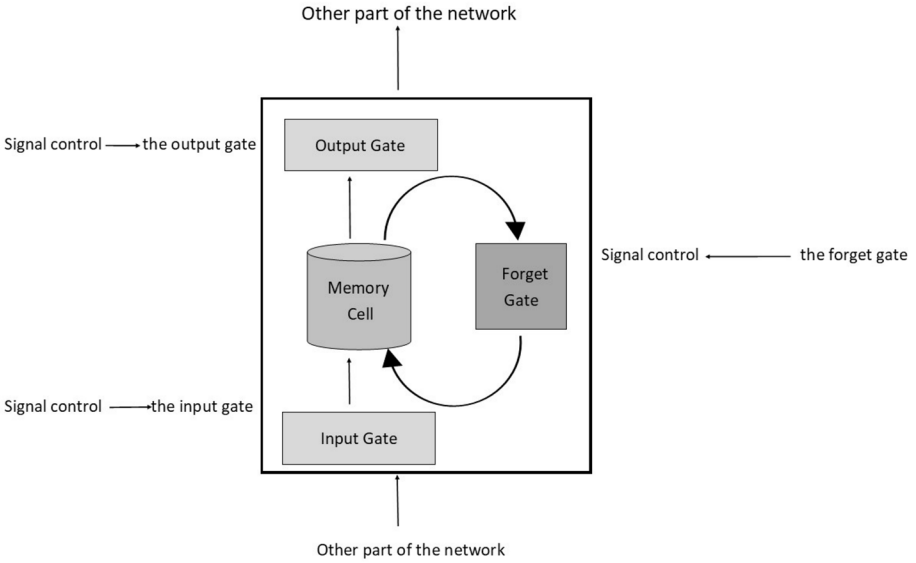


Fig. 5 General architecture of Long Short-Term Memory (LSTM) (Mei et al. 2020)

stake holders in planning water resources more efficiently. The aforementioned qualities also help LSTM in having good generalization ability on data allowing it to predict streamflow accurately on most predictors (Guo et al. 2021b). Specifically, LSTM is known to model parameters that lead streamflow response by large time durations. For example, the impact of snow on streamflow response is observed after months because snow accumulates during winter season and it melts during spring and summer, where it contributes to streamflow. The ability of LSTM to handle long term dependencies in data allows it to model the effect of such parameters accurately (Kratzert et al. 2018).

A recent finding suggests that LSTM could prove to be effective when trained on data from multiple basins and catchments, consequently allowing a single LSTM model to forecast streamflow for multiple entities (Kratzert et al. 2024; Höge et al. 2022). Further research has shown that a single LSTM algorithm is capable of forecasting streamflow over multiple timescales having been trained on data of different resolutions (Gauch et al. 2021). Another prominent feature of LSTM's predictive capability is its ability in forecasting extreme streamflow events without being explicitly trained on extreme events data (Frame et al. 2022). This is another proof of LSTM's enhanced ability to generalize on data (Nearing et al. 2020, 2021; Ma et al. 2024). A prominent quality of LSTM networks is their ability to characterize predictive uncertainties in streamflow modeling, resulting in robust forecast mechanisms (Li et al. 2021b; Vinokić et al. 2025). Findings from a research indicate that LSTM demonstrates applicability in global discharge modeling which is more evidence of LSTM's great generalization ability (Yang et al. 2023).

2.1.4 Fuzzy Inference Based ANN (Adaptive Neuro Fuzzy Inference System (ANFIS))

ANFIS is a fuzzy mapping technique based on the Tagaki-Sugeno-Kang (TSK) fuzzy inference system (El-Shafie et al. 2007). The model functions by combining backpropagation and

least mean square optimization algorithms. ANFIS is fundamentally a graphical network representation of sugeno type fuzzy models. The network consists of nodes and/or duties with specific functions (Terzi and Ergin 2014). Figure 6 depicts the general architecture of ANFIS.

ANFIS has widely been applied in hydrological applications owing to its ability of modeling complex and non-linear relationships amongst variables. This feature of ANFIS has allowed it to forecast streamflow with a high degree of accuracy (Meshram et al. 2022).

Another feature of ANFIS is its ability to handle uncertainties in inputs. This allows it to account for unusual occurrences in streamflow data (sudden rise or fall) (El-Shafie et al. 2007; Terzi and Ergin 2014; Huang et al. 2021), making it superior to commonly used deep learning approaches such as the feed forward ANN that was discussed earlier.

2.1.5 Extreme Learning Machine (ELM) Algorithm

ELM is a state-of-the-art AI algorithm constructed under single layer feed forward ANN. ELM does not require parameters' tuning apart from the pre-defined network architecture. Moreover, the model avoids problems faced by gradient based algorithms such as the feed forward ANN, which include slow learning etc (Deo and Şahin 2016). ELM has thus, become a popular algorithm amongst researchers in the field of hydrology. Figure 7 depicts the general architecture of ELM.

ELM is very apt at learning trends within data which enables it to understand the changing behavior of streamflow in different circumstances. This quality makes it superior to feed forward ANN when forecasting streamflow especially at one time step ahead (Deo and Şahin 2016; Yaseen et al. 2018; Sulugodu and Deka 2019).

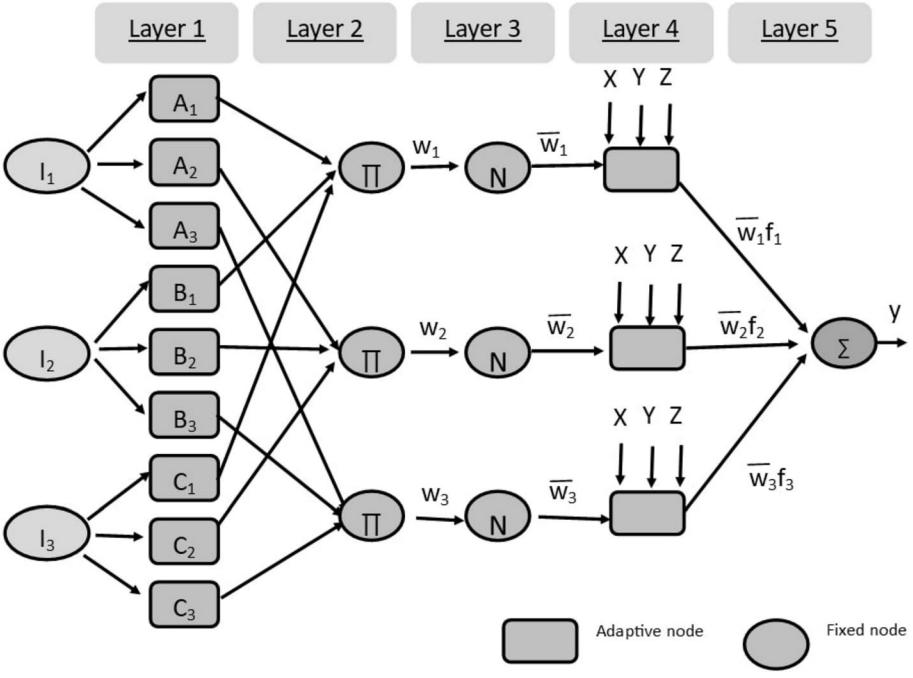


Fig. 6 General architecture of Adaptive Neuro Fuzzy Inference System (ANFIS) (Vafaeenezhad et al. 2014)

A distinct advantage of ELM over some of the other algorithms such as SVM is its fast computational time which makes it very feasible to use in real time streamflow forecasting problems (Yaseen et al. 2016b).

However, ELM is known to struggle when forecasting peak flow values (Niu and Feng 2021). This could be down to its simple structure that deprives it of generalizing on data adequately.

A potential improvement that could be made to ELM is its ability to forecast streamflow multiple time steps ahead. Current research shows that ELM has only been applied in one time step ahead forecasting. The incorporation of feed back or recurrent loops can address this issue.

2.1.6 Bayesian Neural Network (BNN)

The conventional feed forward ANN is known to struggle with over fitting, a phenomenon in which the algorithm performs well when predicting on training data but struggles when its predicting on data outside the training set. The Bayesian approach overcomes this problem as it represents the uncertainty about the relationship between input–output using a probability density function of the parameters. Before data observation, the parameters are described by a prior probability density function. Once the data are observed or collected, the corresponding posterior probability density function can be derived using Bayes' theorem (Muluye and Coulibaly 2007).

The aforementioned features enable BNN to capture useful information from important predictors resulting in enhanced streamflow forecasting accuracy (Muluye and Coulibaly 2007).

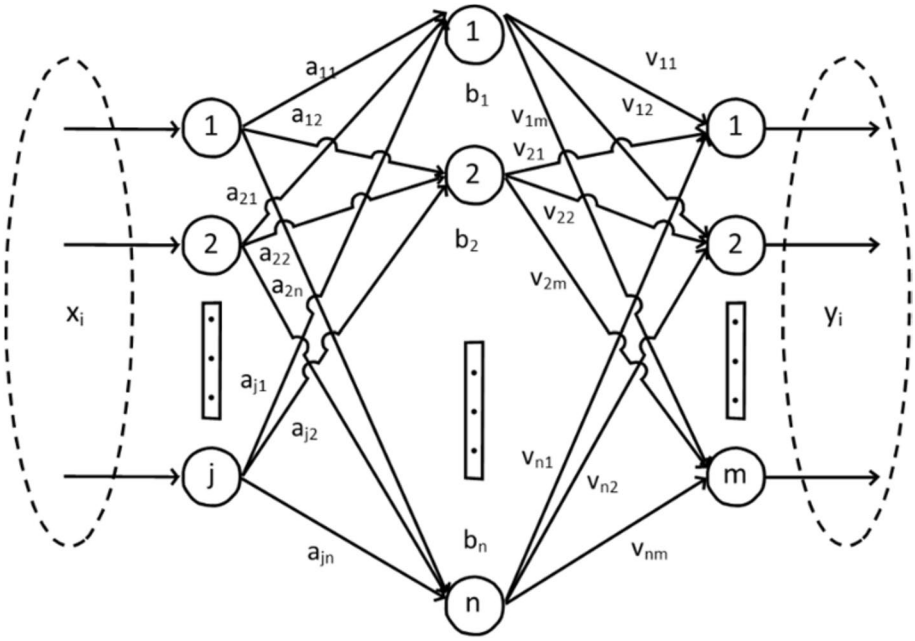


Fig. 7 General architecture of Extreme Learning Machine (ELM) Algorithm (Huang et al. 2011)

A key feature of BNN is its ability to determine uncertainty in hydrological modeling, a feature that is missing in deterministic AI models (Zhang et al. 2009). This aids in robust decision making with regards to water resources management.

2.2 SVM Family

2.2.1 Support Vector Machine (SVM)

SVM comes in two variants, regression-based and classification-based. For streamflow prediction problems, the regression-based variant is used more often. SVM regression makes an estimate of a non-linear function that maps the inputs to output (Adnan et al. 2017). Because of this trait, SVM is one of the most widely used algorithms in streamflow prediction. Figure 8 depicts the general architecture of SVM.

SVM is a very popularly used machine learning technique in regression based forecasting. The fact that SVM is not sensitive to outliers in data makes it a superior method in streamflow forecasting as compared to its counterparts such as the feed forward ANN. Various case studies have reported SVM to perform better than feed forward ANN especially in monthly streamflow forecasting (Adnan et al. 2017; Guo et al. 2011; Kalra and Ahmad 2009; Parisouj et al. 2020).

Since SVM is not overly affected by outliers in data, it forecasts peak streamflow values reasonably accurately (Wu et al. 2009; Karran et al. 2014; Sivapragasam and Liong 2005). This makes SVM a very popular method in contemporary streamflow forecasting research.

2.2.2 Least Squares Support Vector Machine (LS-SVM)

LS-SVM is a type of SVM where the least square method is employed for finding the hyper planes that have the maximum distance from the nearest observed data on both sides. LS-SVM applies the principle of structural risk minimization (SRM) to find out patterns

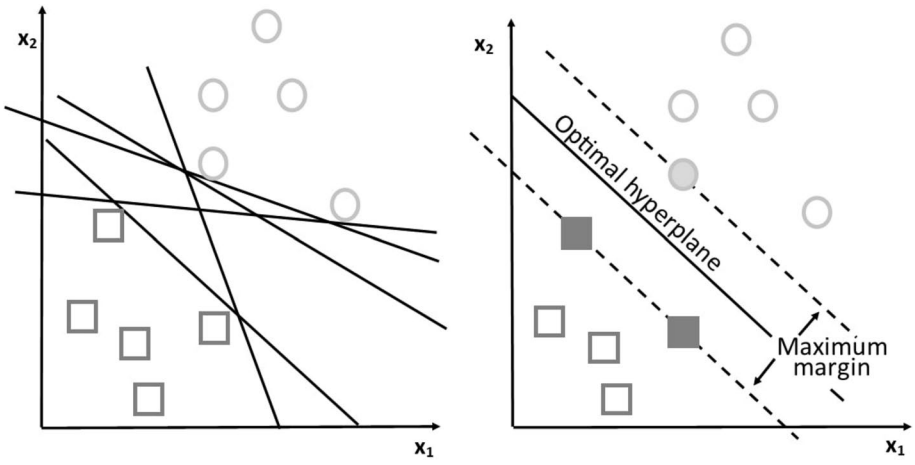


Fig. 8 General architecture of Support Vector Machine (SVM) (Pisner and Schnyer 2020)

between inputs and output (Modaresi et al. 2018). LS-SVM performs non-linear mapping of both the input and output spaces. This feature makes it very useful for streamflow forecasting applications. Figure 9 depicts the general architecture of LS-SVM.

The ability of LS-SVM to model handle non-linearities in data makes it a useful method in streamflow forecasting problems (Modaresi et al. 2018; Zhang et al. 2018b; Yaseen et al. 2016c; Kalteh 2016). This helps it in modeling complex relationships within the given data, thus allowing it to forecast streamflow with a reasonable degree of accuracy.

2.3 Regression Tree Family

Regression trees divide the data into subsets, that is, branches, nodes, and leaves. Regression trees select chunks that decrease the dispersion of target attribute values. Thus, the target attribute values can be predicted from their mean values in the leaves (Loh 2011). Prominent members of the regression tree family include decision tree (DT), random forest (RF) and M5 model tree.

2.3.1 Random Forest

The RF model consists of a combination of randomized regression trees that function by constructing a number of regression trees and then aggregating them to yield a single prediction. The advantages that RF model carries are its inherent feature selection ability which allows it to filter out redundant inputs from training data and its insensitivity to outliers which makes it a useful method when modeling extreme flows (Ghorbani et al. 2020; Tofiq et al. 2022). These features make it considerably useful in streamflow forecasting.

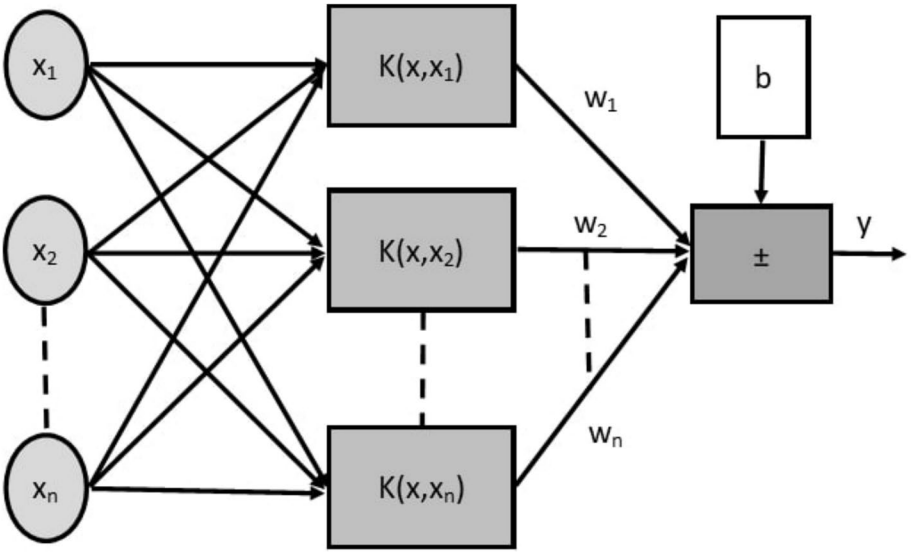


Fig. 9 General architecture of Least Squares Support Vector Machine (LS-SVM) (Okkan and Serbes 2012)

2.3.2 M5 Model Tree

M5 model tree consists of linear functions at the terminal (leaf) nodes, which develops a relationship between inputs and output variable. It operates by splitting Input/output data into subsets to create a decision tree. M5 model tree has proven to be effective in multi-step ahead streamflow forecasting (Yin et al. 2018; Taghi Sattari et al. 2013).

However, M5 model tree has been seen to struggle when modeling peak streamflow (Yin et al. 2018; Yaseen et al. 2016c). A drawback that could be down to sensitivity to outliers in data.

2.4 Hybrid Models

Hybrid AI models refer to a combination of more than one AI models, AI model(s) with evolutionary algorithm(s), AI model(s) with data pre-processing algorithms or AI model(s) with statistical model(s). They are particularly useful in modelling complex problems, where they help in breaking complex tasks into smaller chunks and solve them using individual models. These qualities have led to a surge in the number of studies which have employed hybrid AI models for streamflow forecasting, thus forming a very important part of the techniques used for streamflow modeling.

Hybrid wavelet de-noising and wavelet corrector based AI methods are very popular in contemporary streamflow forecasting research. These methods capture useful information from predictor variables that helps them understand underlying complex patterns in data, thereby allowing them to achieve high accuracy with streamflow forecasts (Hsu et al. 2002; Apaydin et al. 2021; Nayak et al. 2007). This feature also makes them immune to sudden changes in streamflow that could be due to external unexpected events (Ni et al. 2020; Chiang et al. 2018; Chang and Chen 2001; Sanikhani and Kisi 2012; Huangpeng et al. 2021; Rezaie-Balf et al. 2019a; Zhang et al. 2018b).

Evolutionary algorithms as well as wavelet de-noising methods coupled with AI methods result in hybrid systems that are particularly efficient in input variables' selection for a given streamflow forecasting problem. The evolutionary algorithms are apt at feature selection allowing for the removal of redundant inputs and selection of relevant input data for the forecasting problem (Mehr et al. 2013; Osman et al. 2020; Guven 2009). The AI methods perform streamflow modeling using the input data selected based on the decisions made by the evolutionary algorithm. This mechanism leads to efficient forecast systems, yielding a high degree of accuracy as is evidenced by existing literature (Sammen et al. 2021; Feng et al. 2022; Hu et al. 2020; Jain and Kumar 2007; Whigham and Crapper 2001; Taormina and Chau 2015; Kilinc and Haznedar 2022; Danandeh Mehr et al. 2022; Hassan and Hassan 2021; Kalteh 2016; Nourani et al. 2017).

The combination of two AI methods tend to yield very strong predictive systems. Specifically, the fusion of deep learning methods specialized in solving tasks of different nature resultant in predictive models that are very robust. Researchers have sparingly hybridized deep learning methods specialized in information extraction with those able to model sequential data and have been successful in yielding good forecast results (Ghimire et al. 2021; Feng et al. 2020a; Danesh et al. 2025).

Hybrid AI methods including wavelet based hybrid AI models, those formed by fusion of evolutionary algorithms with AI methods, fuzzy logic based hybrid AI techniques, data

pre-processing coupled AI methods and fusion of two AI methods etc all tend to generalize very well on non-linear streamflow and other predictors' data. This enables them to forecast peak high and low flows with a considerable degree of accuracy (Qian et al. 2025; Wang et al. 2026; Nayak et al. 2007; Yaseen et al. 2017; Guo et al. 2011; Dariane and Azimi 2018; Sahay and Srivastava 2014; Dhanya and Nagesh Kumar 2011; Ahmed et al. 2022; Liu et al. 2020; Badrzadeh et al. 2013; Rezaie-Balf et al. 2019b; Wang et al. 2023; Adamowski and Sun 2010; Zhang et al. 2018a; Danandeh Mehr et al. 2014; Pramanik et al. 2011; Makwana and Tiwari 2014; Mohammadi et al. 2021; Nacar et al. 2018; Mehdizadeh et al. 2019; Araghinejad et al. 2018; Özger 2009). This is largely down to individual models in the hybrid system solving problems for which they are specialized.

3 Evaluation Metrics

Streamflow forecast systems are evaluated using different skill metrics. Each has its own pros and cons largely dependent on the nature of the problem. The mean absolute error (MAE) represents the average of the absolute difference between the actual and predicted values in the dataset. Equation 1 represents MAE mathematically.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - y_i^{'}| \tag{1}$$

where y_i and $y_i^{'}$ represent the actual and predicted values respectively. MAE measures the average of the residuals in the dataset. It is advantageous in that it is simple and fast to compute. It has the same unit as the variable being modelled, making it easy to interpret. MAE has its disadvantages too. For instance, MAE can be dominated by even one outlier or large error (Onyutha 2024).

The root mean squared error (RMSE) is the square root of mean squared error. It measures the standard deviation of residuals and is represented by Eq. 2.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - y_i^{'})^2} \tag{2}$$

where y_i and $y_i^{'}$ represent the actual and predicted values respectively.

RMSE carries several advantages. It is fast to compute and holds the same unit as the variable being modelled, making it easy to interpret. Just like MAE, its disadvantage is that it can be affected by the presence of outliers in the data (Onyutha 2024).

The correlation coefficient (R) represents the degree of association between two variables. It is a frequently used metric in different forecasting problems. It is represented by Eq. 3.

$$R = \frac{\sum_{i=1}^{n}(y_i^{'} - y_i^{'^-})(y_i - y^-)}{\sqrt{\sum_{i=1}^{n}(y_i - y_i^{'^-})^2 \sum_{i=1}^{n}(y_i - y^-)^2]}} \tag{3}$$

R is advantageous in that it is easy to compute but has a major disadvantage of assuming linear relation between the variables under question (Asuero et al. 2006).

The coefficient of determination (R^2) represents the degree of the variance in the target variable which is explained by the linear regression model. It is represented by Eq. 4.

$$R^2 = 1 - \frac{1}{n}\frac{\sum_{i=1}^{n}(y_i - y_i^{'})^2}{\sum_{i=1}^{n}(y_i - y_i^{-})^2} \tag{4}$$

where y_i, $y_i^{'}$ and y_i^{-} represent the actual, predicted and mean of actual values respectively.

Advantages ofR^2 are that it can be computed fast, and does not have interpretability limitations. Interpretation of R^2 is straightforward in terms of the percentage of the variability explained by the model. R^2 is disadvantageous in several aspects. It is sensitive to outliers and is unable to quantify bias. The most commonly used formula for R^2 is not suitable for analysis of non-linear relationships (Onyutha 2024).

The Nash-Sutcliffe Efficiency (NSE) is defined as the one minus the ratio of the error variance of the modeled time-series divided by the variance of the observed time-series. It is a commonly used metric for evaluating performances of hydrological forecast models. It is mathematically represented by Eq. 5.

$$NSE = 1 - \frac{\sum_{t=1}^{T}(Q_o^t - Q_t^m)^2}{\sum_{t=1}^{T}(Q_o^t - Q_t^-)^2} \tag{5}$$

where Q_t^- is the mean of observed discharges, Q_t^m is modeled discharge and Q_o^t is observed discharge at time t.

NSE holds advantages pertaining to its simplicity and fast computational time. NSE has its disadvantages also. It has limitations pertaining to its interpretability as compared to other metrics which vary from zero to one. Moreover, NSE lacks physical meaning since it does not have a unit. NSE can be sensitive to outliers. The sampling uncertainty in NSE is quite large. The dependence of NSE on average of observations leads to exaggerated model efficiency when analysing highly seasonal river flow (Onyutha 2024).

Apart from the above mentioned skill metrics, some other used metrics include the squared correlation coefficient r^2, Willmott index (WI), normalized root mean squared error (NRMSE) and mean absolute relative error (MARE). Figure 10 depicts the number of studies that have used each evaluation metric or a combination of metrics for evaluating streamflow forecasting models.

RMSE has been most commonly employed by researchers in evaluating streamflow forecasting systems, as is evident from Fig. 10. NSE has also been frequently used as it has specifically been developed for evaluating hydrological problems. R^2, R and MAE because of their fast computation and simple implementation have also been popular amongst researchers.

4 Analysis and Evaluation

This paper has conducted a comprehensive study on the various AI methods applied for streamflow forecasting. Several papers are reviewed for analysing their effectiveness in modelling streamflow over different lead times.

Most of the studies have employed feed forward ANN models for streamflow forecasting, as is evident from Fig. 11. Adding to this, hybrid models have proven to be very effective

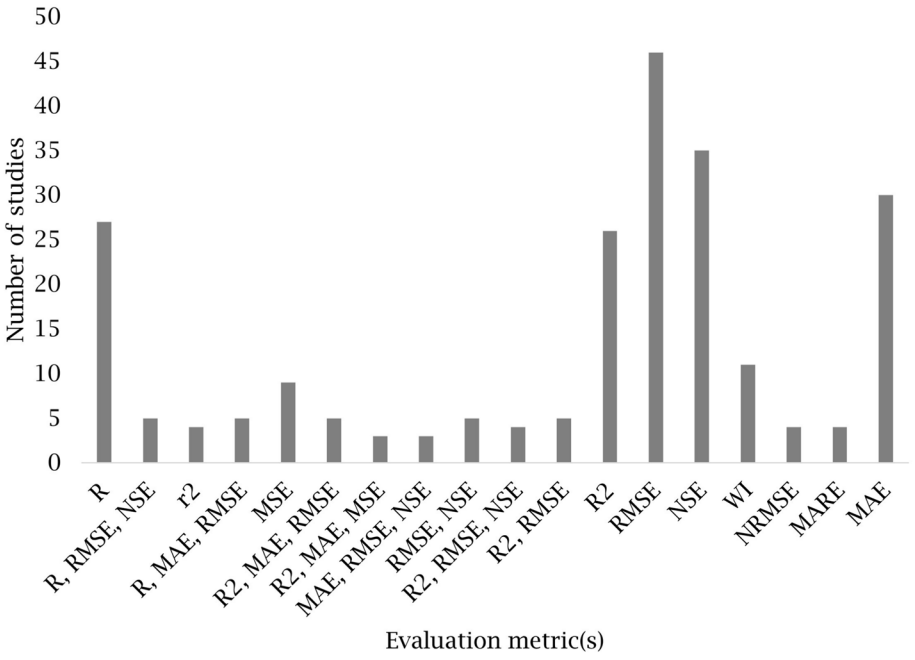


Fig. 10 Number of studies that have used each evaluation metric or a combination of skill metrics

in improving the forecast accuracies of standalone models. The studies on hybrid models for streamflow forecasting have been dominated by the ones built by combining an optimization algorithm with an AI model. The optimization algorithm is used either for pre-processing time series data or for selecting appropriate input features, while the AI model makes streamflow predictions. ANN has been most commonly used as it is best suited to modeling non-linear data, giving it a strong generalization capability. Neural networks have formed the basis for further improvements in streamflow modeling, the most significant one being the introduction of recurrent networks which have immensely improved forecast results of variables varying with time.

The examined studies have forecasted streamflow over different lead times. As can be seen in Fig. 12, studies with 1-7 day(s) lead time exceeds in number in comparison to other forecast time frames. The reluctance on part of the researchers to experiment with larger lead times i.e., 2 weeks or beyond could be due to an overall decrease in forecasting accuracy over large lead times. Apart from the 1–7-day(s) lead time horizon, researchers have frequently employed 1 month ahead streamflow forecasts as well. But forecasts for this lead time have been performed using monthly resolution data, which makes the 1-month lead time a short forecast window in this context. Researchers have used high resolution data, i.e., hourly data, sparingly for streamflow forecasting. This could be due to lack of availability of such data for most locations. However, forecasting streamflow on such high resolution data could potentially give stake holders more leverage in terms of time available for managing water resources.

Selection of the best input variables' combination has been a challenge for most researchers in the reviewed literature. Not only the variables but the selection of lag size of variables (number of previous values to look at) for predicting streamflow is a parameter whose optimum value could vary from case to case. Resolution of data has a major say in the optimal

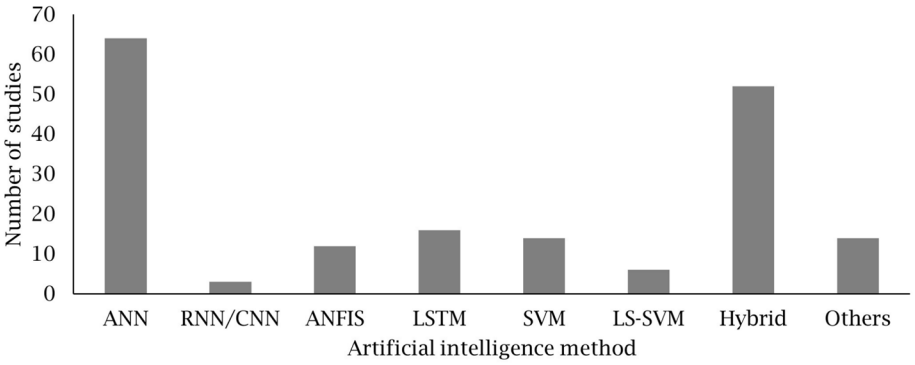


Fig. 11 Number of studies conducted on each Artificial Intelligence (AI) method

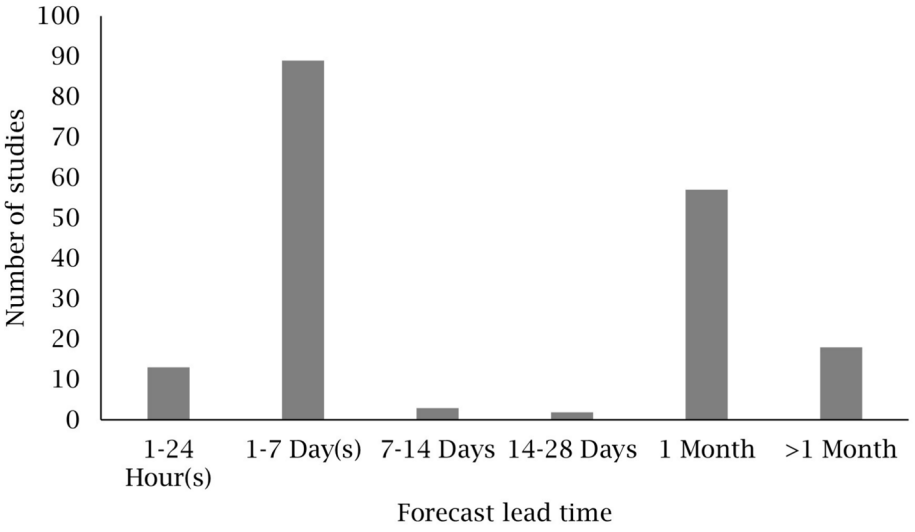


Fig. 12 Number of studies conducted on each forecast lead time

lag size of input variables. Researchers have used different input variables' combinations in the various studies referred to in this paper.

Most studies have employed univariate models where only lagged streamflow values have been used as model inputs. This is evident in Fig. 13. Some other combinations including streamflow with rainfall as well as streamflow, rainfall and temperature together have also been used frequently for streamflow forecasting. Correlation tests have been used in different studies to find the optimal input variables' combination for a specific case. The choice of input variables' combination also depends on the available data for a specific region. For instance, snow is a direct contributor to streamflow, but snow data is not readily available for most locations. Hence, snow parameters have been less commonly used in streamflow prediction problems.

The use of AI methods as standalone models as well as in combination with other algorithms (hybrid models) have witnessed great increase with each passing year. Figure 14 shows the

number of studies which have used different AI models for streamflow forecasting each year spanning from 1999 to 2023. The use of AI methods for streamflow forecasting have generally increased over subsequent years, with ANN being the most commonly used model. As discussed earlier, this is because of the ability of ANNs to handle non-linearities in data and generalize on data in a better way as compared to other models. Hybrid models have also attracted attention of researchers as they are apt at solving multiple problems in the forecasting paradigm.

Figure 15 depicts the comparative visualization of the number of studies that have applied different input variables for each AI method used. As can be seen in this figure, antecedent streamflow values have been employed most commonly with hybrid models followed by ANN models. Streamflow and rainfall in combination come next, which again are employed most in hybrid, followed by ANN models. This shows that streamflow used in a univariate setting yields satisfactory forecast results as well as requiring less computational resources. Moreover, with the optimal selection of input variables, researchers have most frequently employed hybrid models as they help in solving complex problems simultaneously with specialized algorithms.

Figure 16 depicts a comparison of the number of studies that have applied different forecast lead times for each AI method. Forecast lead times of 1-7 day(s) and 1 month are most commonly applied with ANN and hybrid models. It is evident that researchers have been inclined towards forecasting streamflow over a maximum of 7 days and 1 month lead time, as data for daily and monthly resolution is readily available as well as these forecast horizons being less demanding on the predictive system.

Figure 17 shows the number of studies that have employed different input variables for each forecast lead time used. It can be seen that antecedent streamflow and combination of streamflow and rainfall as model inputs have most commonly been used with lead times of 1-7 day(s) and 1 month. The reviewed literature suggests that there is a lack of studies on the usage of time series models for streamflow modelling. Figure 11 suggests that both RNN

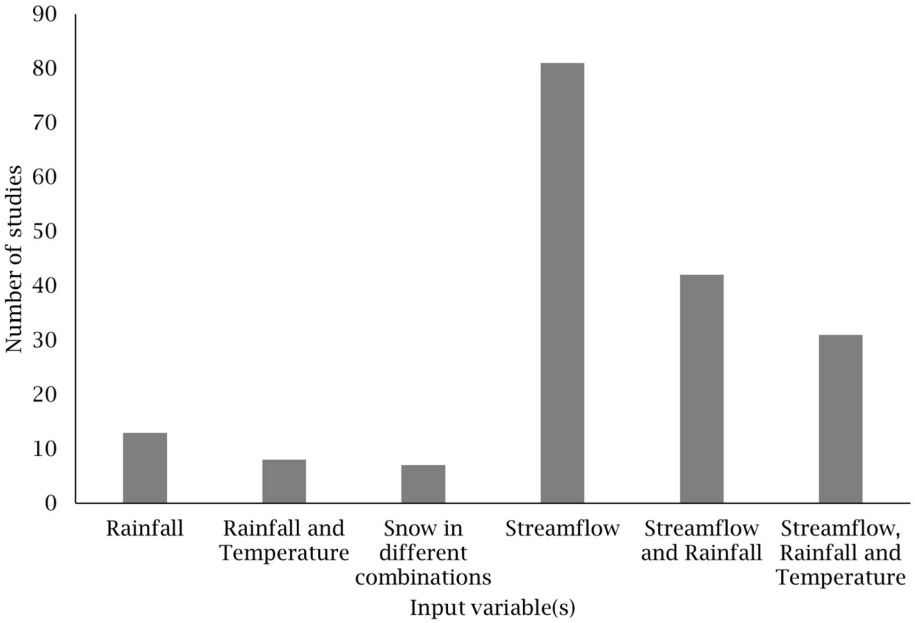


Fig. 13 Number of studies on different input parameter combinations

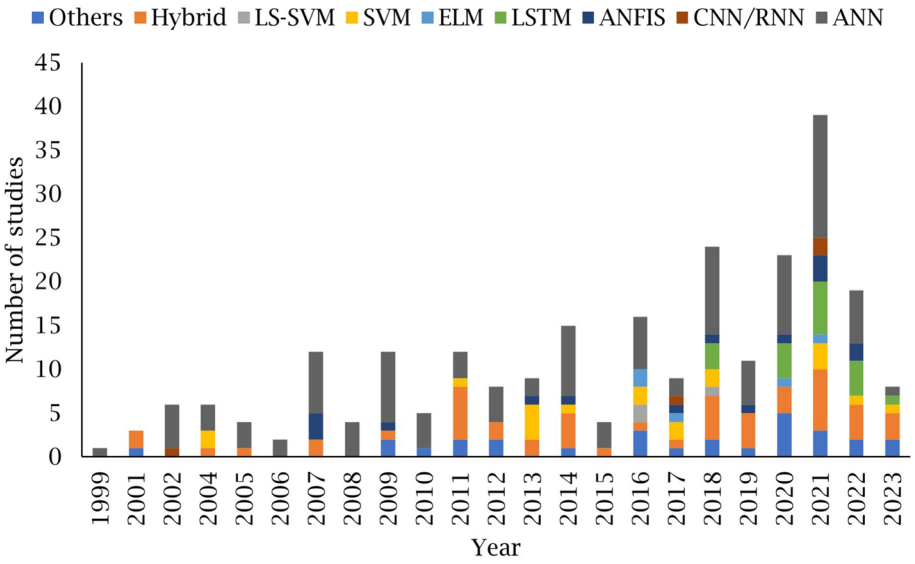


Fig. 14 The use of different AI methods for streamflow forecasting in each year

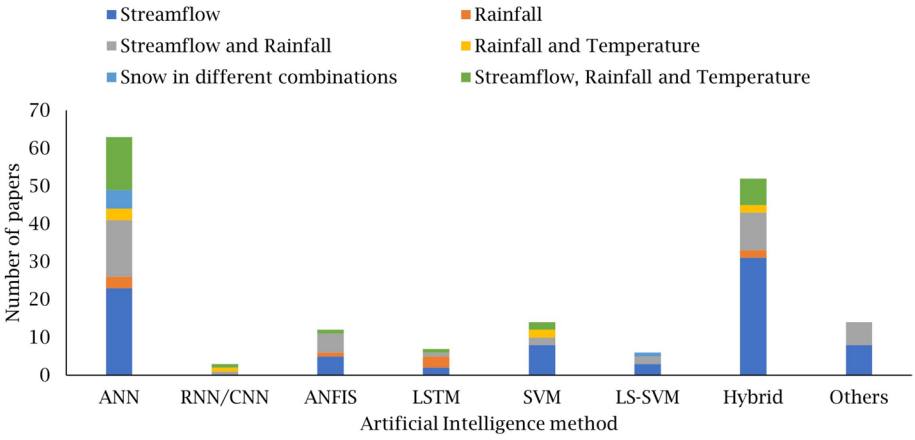


Fig. 15 The number of studies on the use of different input variables for each AI method used

and LSTM have been sparingly used for streamflow forecasting when they should ideally be employed more often. Both these models contain feedback mechanisms which could enhance forecast accuracies. An extensive effort must be made in developing models that could generate forecasts over large lead times. In the reviewed literature, most studies have performed forecasts for 1–7-day(s) lead time. This lead time has potential to be increased, especially for cases which involve planning of water resources for different applications.

While most studies have used streamflow, rainfall and temperature in different combinations as input variables, there is scope for investigating the effects of additional predictors such as snow and wind speed etc. Snow is a big contributor to streamflow but has not been

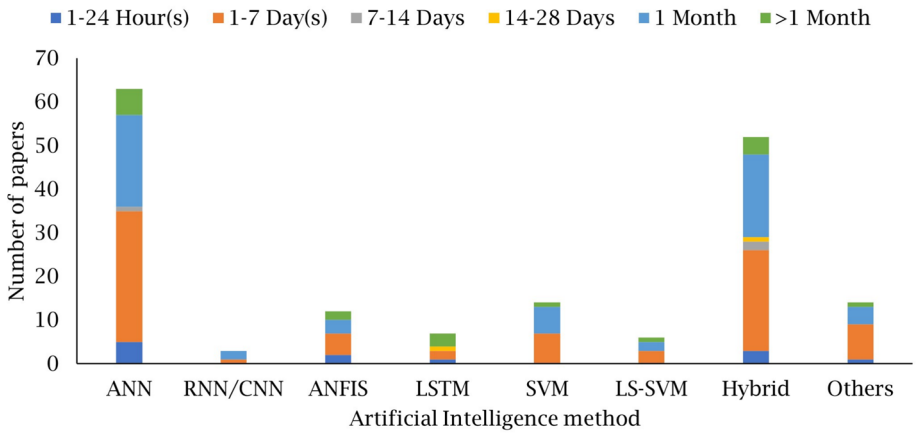


Fig. 16 The number of studies on the application of different forecast lead times for each AI method used

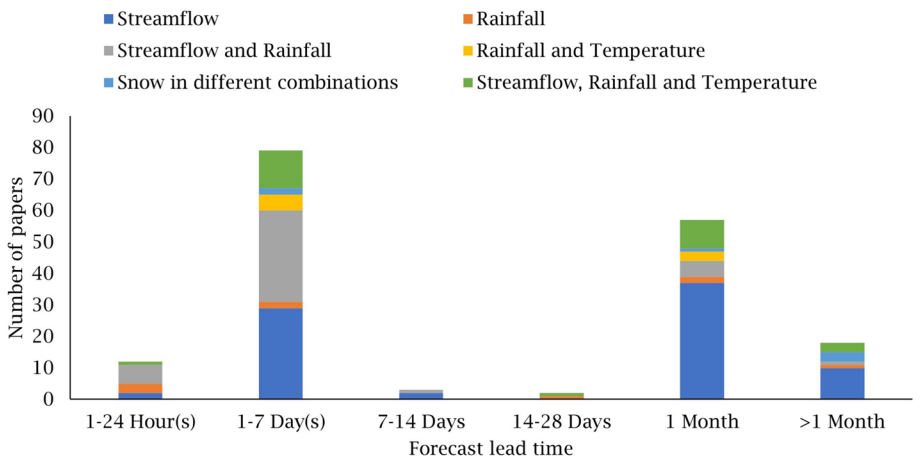


Fig. 17 The number of studies on the use of different input variables for each forecast lead time applied

used often in streamflow forecasting. This is because of the lack of snow data available for most locations. It is, thus, important to use snow as a predictor if snow data is available for a specific location. Researchers also tend to underestimate the effect of wind speed on streamflow which is evident from the fact that wind speed has hardly been used as an input variable for streamflow prediction by researchers.

5 Recommendations for Future Work

The past decade has witnessed a huge surge in the application of AI techniques for streamflow forecasting. Researchers have tried to address the numerous challenges that have come along the way when forecasting streamflow. There, however, remain a few aspects that need improvements.

1. Most studies on streamflow forecasting have been performed using streamflow, rainfall and temperature in different combinations as input parameters. There remains room for more exploration with regards to this aspect. Some other variables like snow and wind-speed seem under represented in AI studies, and may prove useful to improve streamflow forecasting. Snow is a direct contributor to streamflow while wind speed is known to impact it indirectly, as wind can be attributed to weather systems that bring rain. The use of both these parameters and other potential parameters could improve forecast accuracy and reliability of AI models for streamflow forecasting.
2. Forecast window size is an important facet of any prediction problem. Researchers have generally employed a lead time of up-to 7 days on daily resolution data, which makes it a forecast window of up to seven time steps. An increase in forecast window results in a decrease in forecast accuracy in general. For planning purposes, larger lead times are desirable, since they give stakeholders a better opportunity for preparation. There is hence, a need for developing a robust forecasting model that could give forecasts over larger forecast windows without compromising on forecast accuracy.
3. As far as neural networks are concerned, researchers have predominantly employed feed forward ANN for streamflow forecasting. Very few studies have been performed on time series neural network models such as RNN and LSTM. Both these methods could be effective in modelling complex data used for streamflow forecasting. LSTM, in particular, could be very effective in generating forecasts over larger windows. This intuition is based on LSTM's ability of maintaining long term memory which helps it in developing context for subsequent forecasts. Hence, more studies need to be done on time series neural network models for streamflow forecasting.
4. Very little research has been performed on high resolution data, for example, hourly time step data when forecasting streamflow. As far as the studies that have used hourly time step data are concerned, only few time steps ahead have been forecasted. Multi-days ahead forecasting have not yet been researched on when using hourly time step data. The advantage of using hourly time step data is that useful information is not lost as opposed to when using data of lower resolution. However, forecasting multi-days ahead streamflow on hourly time step data is a huge challenge even in modern research as the number of forecast time steps get very large. For example, one week ahead forecasts on hourly time step data means forecasting over 168 time steps. Forecasting over so many time steps has adverse effects on forecast accuracy. The key is to develop robust mechanisms that can forecast streamflow over so many time steps without letting the accuracy of forecasts falling below a certain acceptable level. This will be key milestone in modern research on streamflow forecasting as it will allow for much better planning of water resources.
5. AI has the potential to substantially reduce the computational time required for real-time flood inundation prediction. This is important for flash floods, which can develop rapidly, following intense rainfall and provide only a limited window for emergency response. Conventional hydrological models can require considerable computational time, especially when high resolution simulations are needed to predict the spatial extent and depth of inundation. Once trained using historical rainfall, streamflow, topographic and inundation data, AI-based models can estimate flood characteristics and potentially provide near real-time predictions. This computational efficiency could support more timely flood warnings and emergency decision making, particularly in data-rich environments where rapid prediction of potentially affected areas is critical.

6 Conclusion

This review article analysed and summarized studies on the application of AI methods for streamflow forecasting over a 25 year period i.e., 1999-2023. AI methods have aided considerable advancements in streamflow forecasting problems owing to their ability of handling non-linearity, non-stationary and complexity in datasets. The reviewed literature suggests that the most significant aspect that affects the accuracy of an AI model is the input parameter combination. Of the various input parameter combinations, the most influential parameter was historical streamflow. The next most influential parameter was current and historical rainfall followed by temperature. For a relatively limited number of studies, snow was also a key predictor, as it is a direct contributor to streamflow. These input parameters have been used with success through past studies. However, there are many parameters that have not yet been investigated, and that could potentially enhance streamflow forecasting accuracy.

A crucial facet in the selection of input parameter combination is the appropriate choice of lag size (number of time steps to look into the past for each input) for the inputs. Reviewed literature suggests that a lag of 2-4 days is near optimal for streamflow, rainfall and temperature when used as input variables for forecasting streamflow. However, this may not hold true for other data resolutions. For instance, when predicting streamflow at hourly time step, the optimal lag size for input variables may be different from the one specified for forecasting streamflow on daily time step data and will need to be determined on a case by case basis. The review suggests that different input parameters may need individual configurations to provide their best predictive capability. Literature did not reveal any systematic consideration for the individual configuration of input parameters, and this remains an area for further research and exploration. Furthermore, each stream has a unique topography, geology, weather, vegetation and ground cover, land use type and historical changes, which makes the selection of lag size for each parameter a challenging but important task.

The literature also revealed that as the forecast window is increased model accuracy generally reduces due to the stochastic nature of streamflow, and its predictors. For operational purposes, short term forecasts of hours or just a few days into the future are performed where model performance does not deteriorate much as the short forecast lead time does not test the limit of the model completely. For planning purposes however, long term forecasts spanning over months are performed. For such large prediction horizons, recursive forecasting is employed, wherein each input variable is forecasted at every time step and the resultant value is fed into the model as input along with forecasted value of target variable (streamflow in this case) for generating the next forecast. This mechanism ultimately reaches a stage where the model's capability of forecasting streamflow with high degree of accuracy is completely tested and its forecast accuracy starts to decrease very quickly over successive forecasts.

AI techniques, as evidenced by the substantial literature, are at the heart of current research in streamflow forecasting. ANNs in different forms are currently being employed most frequently for streamflow forecasting. Hybrid models are also very popular amongst current researchers. However, there lies a lot of room for improvement, especially in long term streamflow forecasting (more than two weeks ahead), which is currently the biggest challenge in streamflow forecasting. Future studies, for instance, could consider employing robust time series models having the ability of generating forecasts over larger windows without compromising much on forecast accuracy. There also lies room for experimenting with more input variables in addition to the already tried and tested ones, i.e., streamflow,

rainfall and temperature. Wind speed seems to be an under-utilised parameter as evidenced by extensive literature review. It has the potential to influence rainfall and might add useful information as an input variable to streamflow forecasting problems.

Appendix A: AI Methods Used in Literature

Table 1 AI methods used in literature

Acronym	Used in
ANN family	
FFANN	Chitsaz et al. (2016); Selvi and Huseyinov (2020); Zheng et al. (2018); Cutore et al. (2007); Dolling and Varas (2002); Sharma et al. (2021); Besaw et al. (2010); Chokmani et al. (2008); Vidyarthi and Jain (2022); Can et al. (2012); Ahmed and Sarma (2007); Toth and Brath (2007); Wu et al. (2005); Fan et al. (2021); Cui et al. (2020); Sun et al. (2019); Muluye and Coulibaly (2007); Worland et al. (2019); Ali and Shahbaz (2020); Birikundavyi et al. (2002); De Faria et al. (2022); Dehghani et al. (2014); Makkeasorn et al. (2008); Sattari et al. (2012); Uysal et al. (2016a); Choudhury et al. (2018); Parasuraman et al. (2006); Gunathilake et al. (2021); Chiang et al. (2004); Hsu et al. (2002); Wang et al. (2019); Hu et al. (2005); Danandeh Mehr et al. (2015); Moradkhani et al. (2004); Cigizoglu (2005); Yaseen et al. (2016a); Sharma et al. (2022)
Less common ANNs (CNN, RNN)	Apaydin et al. (2021); Shu et al. (2021); Chiang et al. (2017)
LSTM	Apaydin et al. (2021); Huang et al. (2021); Asaad et al. (2022); Cheng et al. (2020); Wang et al. (2022); Kratzert et al. (2018, 2024); Gauch et al. (2021); Frame et al. (2022); Nearing et al. (2020, 2021); Ma et al. (2024); Li et al. (2021b); Yang et al. (2023); Höge et al. (2022)
ANFIS	Meshram et al. (2022); El-Shafie et al. (2007); Terzi and Ergin (2014); Vatanchi et al. (2023)
ELM	Deo and Şahin (2016); Sulugodu and Deka (2019); Yaseen et al. (2016b, 2018, 2019); Niu and Feng (2021); Liu et al. (2023)
BNN	Muluye and Coulibaly (2007); Zhang et al. (2009)
SVM family	
SVM	Adnan et al. (2017); Guo et al. (2011); Kalra and Ahmad (2009); Parisouj et al. (2020); Wu et al. (2009); Karran et al. (2014); Sivapragasam and Liong (2005)
LS-SVM	Modaresi et al. (2018); Zhang et al. (2018b); Yaseen et al. (2016c); Kalteh (2016)
Regression tree family	
RF	Ghorbani et al. (2020); Tofiq et al. (2022)
M5 model tree	Yin et al. (2018); Taghi Sattari et al. (2013); Yaseen et al. (2016c)
Hybrid models	Hsu et al. (2002); Apaydin et al. (2021); Nayak et al. (2007); Ni et al. (2020); Chiang et al. (2018); Chang and Chen (2001); Sanikhani and Kisi (2012); Huangpeng et al. (2021); Rezaie-Balf et al. (2019a); Zhang et al. (2018b); Mehr et al. (2013); Osman et al. (2020); Guven (2009); Sammen et al. (2021); Feng et al. (2022); Hu et al. (2020); Jain and Kumar (2007); Whigham and Crapper (2001); Taormina and Chau (2015); Kilinc and Haznedar (2022); Danandeh Mehr et al. (2014, 2022); Hassan and Hassan (2021); Kalteh (2016); Nourani et al. (2017); Ghimire et al. (2021); Feng et al. (2020a); Yaseen et al. (2017); Guo et al. (2011); Dariane and Azimi (2018); Sahay and Srivastava (2014); Dhanya and Nagesh Kumar (2011); Ahmed et al. (2022); Liu et al. (2020); Badrzadeh et al. (2013); Rezaie-Balf et al. (2019b); Wang et al. (2023); Adamowski and Sun (2010); Zhang et al. (2018a); Pramanik et al. (2011); Makwana and Tiwari (2014); Mohammadi et al. (2021); Nacar et al. (2018); Mehdizadeh et al. (2019); Araghinejad et al. (2018); Özger (2009); Zhou et al. (2023)

Acknowledgements This research is supported by an Australian Government Research Training Program (RTP) Scholarship and is co-funded by Snowy Hydro Pty Ltd.

Author Contributions S. Eswara, A. Barton, T. Choudhury and T. Chubb all contributed to the study conception and design. Data was provided by T. Chubb from Snowy Hydro. Data processing and modelling was performed by S. Eswara. The first draft of the manuscript was written by S. Eswara and S. Eswara, A. Barton, T. Choudhury and T. Chubb all involved with subsequent reviews and improvements to the manuscript. S. Eswara, A. Barton, T. Choudhury and T. Chubb have read and approved the final manuscript.

Funding Open Access funding enabled and organized by CAUL and its Member Institutions. This research is supported by an Australian Government Research Training Program (RTP) Scholarship and is co-funded by Snowy Hydro Pty Ltd.

Availability of Data and Materials Federation University provided computing hardware and software, while Snowy Hydro Pty Ltd provided the relevant data.

Declarations

Ethical Approval Ethics approval was not applicable to this research.

Consent to Participate S. Eswara, A. Barton, T. Choudhury and T. Chubb all consented to their participation.

Consent to Publish S. Eswara, A. Barton, T. Choudhury and T. Chubb all consent to this work being published.

Competing Interests S. Eswara, A. Barton, T. Choudhury and T. Chubb have no relevant financial or non-financial interests to disclose.

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